

Goodness-of-fit test: Testing if population has a specified theoretical distribution.

Eg. Suppose you throw a die 120 times, record the outcome and you observe 1 20 times, 2 22 times, ... 6 24 times.

Expected frequencies with prob. of $1 = 2 = 3 = \dots = 6 = \frac{1}{6}$
Each number $\uparrow$ to 6 20 times

Table 10.4: Observed and Expected Frequencies of 120 Tosses of a Die

Face:	1	2	3	4	5	6
Observed	20	22	17	18	19	24
Expected	20	20	20	20	20	20

Goodness-of-Fit
Test

$$\chi^2 = \sum_{i=1}^k \frac{(o_i - e_i)^2}{e_i}, \sim \chi_{k-1}^2$$

$\chi^2_{\text{small}} \Rightarrow$ good fit

$\chi^2_{\text{big}} \Rightarrow$ bad fit

Each of the expected frequencies must be at least 5.
 $k = \# \text{ terms in the sum}$

Decision $\Rightarrow \chi^2 > \chi_{\alpha, k}^2$, Reject H_0 : Data fits the specified distⁿ

Example: Frequency distribution of battery lives $\sim N(3.5, 0.7^2)$?

Table 10.5: Observed and Expected Frequencies of Battery Lives, Assuming Normality

Class Boundaries	o_i	e_i
1.45-1.95	2	0.5
1.95-2.45	1	2.1
2.45-2.95	4	5.9
2.95-3.45	15	10.3
3.45-3.95	10	10.7
3.95-4.45	5	7.0
4.45-4.95	3	3.5

$$\chi^2 = \frac{\sum (o_i - e_i)^2}{e_i} \sim \chi_3^2$$

Test for independence : We may want to assess if two variables are indpt. or not

H_0 : A & B are independent

H_1 : A & B are not independent.

Contingency table →

Table 10.6: 2×3 Contingency Table

Tax Reform	Income Level			Total
	Low	Medium	High	
For	182	213	203	598
Against	154	138	110	402
Total	336	351	313	1000

$r \text{ rows} = 2$

$c \text{ columns} = 3$

$$(r-1)(c-1) = 2$$

$P(\text{Low} \cap \text{For}) = P(\text{Low}) P(\text{For})$
under H_0

$$= \frac{336}{1000} \cdot \frac{598}{1000} = p_i$$

$$e_i = np_i = 1000 \times p_i = \frac{336 \times 598}{1000}$$

Table 10.7: Observed and Expected Frequencies

Tax Reform	Income Level			Total
	Low	Medium	High	
For	182 (200.9)	213 (209.9)	203 (187.2)	598
Against	154 (135.1)	138 (141.1)	110 (125.8)	402
Total	336	351	313	1000

Test for Independence Calculate

$$\chi^2 = \sum_i \frac{(o_i - e_i)^2}{e_i}, \checkmark$$

where the summation extends over all rc cells in the $r \times c$ contingency table.

If $\chi^2 > \chi^2_{\alpha}$ with $v = (r-1)(c-1)$ degrees of freedom, reject the null hypothesis of independence at the α -level of significance; otherwise, fail to reject the null hypothesis.

Tests for homogeneity:

Suppose now the row sums or column sums are all known in advance

	A	B	C	
X	.	.	.	p_x
Y	.	.	.	p_y
	350	180	200	730

Whether p_x is same for A, B and C?
similarly p_y is same " " " " " ?